

RESEARCH ARTICLE

Investigation of the Application of Machine Learning Algorithms in Credit Risk Assessment of Medium and Micro Enterprises

YUIE ZHAO^{ID}

College of Management, Shenzhen Institute of Information Technology, Shenzhen, Guangdong 518172, China

e-mail: yujie-zhao@hotmail.com

This work was supported by the Annual Project of the 14th Five-Year Plan for Shenzhen Educational Science (The Stakeholder Framework for Implementing Collaboration (SFIC) Collaborative Developing Model of Lifelong Education in Shenzhen under Regular Epidemic Prevention and Control) under Grant dwzz22155–2022.

ABSTRACT Micro and medium enterprises rely on small finance investments and returns to ensure smooth and prolonged functional processes. Due to improper planning and recovery, credit-based investments and lending sometimes lead to financial downfalls. This article analyzes the adverse impacts of credit risk in such enterprises with a solution. This article introduces a Consistent Assessment Method (CAM) using Linear Learning Analysis (LLA) to address the common issues in credit investments and lending. The proposed method identifies the non-recoverable and prolonged investments that cause overheads over the different financial quarters. Identifying such investments/ lending provides knowledge of credit risks over varying market fluctuations. The learning provides two linear analyses: credit increment and financial development. These analyses are performed quarterly, considering the investment and its return consecutively after the awaited period. The issues are analyzed using linear learning that identifies inconsistent financial variations between investment and credit based on fluctuations and non-recoverable instances. Such identification aids in classifying the variants between financial improvement or risks under different quarters for better financial revisions. Thus, this proposed method achieves 14.65% high-risk detection with 5.56% financial revisions for the industry considered by the data source.

INDEX TERMS Credit risk, financial variation, linear learning, medium and micro enterprises.

I. INTRODUCTION

Medium and micro enterprises are widely presented worldwide, providing necessary goods and products to the people. Medium and micro enterprises improve a country's overall economic growth and increase people's economic status [1]. The minimum investment is required to build medium and micro enterprises. Micro enterprises ensure the profit range of the organizations that reduce the latency in the planning process [2]. Credit risk assessment is a process that assesses or addresses risks on payments. Credit risk assessment is a complicated task to perform in medium and micro enterprises [3]. Common issues and financial variations are assessed using a proper assessment process. Various methods are used for the

credit risk assessment process [4]. An enterprise trade credit risk assessment (ETCRA) model is mostly used to identify the risks presented in enterprises. The ETCRA model analyzes the profit and loss of enterprises to detect the exact flow of enterprises. The ETCRA model uses important financial variables to improve the characteristics of the assessment process [5], [6]. A fuzzy cognitive map (FCM) based credit risk assessment method is also used to solve the risks early. The FCM algorithm detects the risks in the early stage, which minimizes the complexity of further evaluation processes [7].

Credit risk analysis is a process that analyzes the risks which are presented in investment. Credit risk is caused by severe enterprise damage, leading to market fluctuations [8]. Investment is a major process that is required for every enterprise. Credit risks mostly occur during investment, causing damage to enterprises [9]. Credit risk analysis is mainly used

The associate editor coordinating the review of this manuscript and approving it for publication was Gang Li^{ID}.

to enhance the performance level of medium and micro enterprises. A neural network (NN) algorithm-based risk analysis method is used in enterprises [10]. The NN algorithm uses a feature selection method to select important features for the risk analysis process. The selected feature contains details such as investment, production level, and financial variation of enterprises [11], [12]. The NN algorithm-based method maximizes the effectiveness ratio of small enterprises. Credit risks are evaluated based on market fluctuations, which minimizes the latency of the risk analysis process [13]. A linear learning analysis technique is used for credit risk analysis. The linear learning technique identifies the issues which are presented in investments. The linear learning technique minimizes the complexity of the risk analysis process using market fluctuations [14].

Machine learning (ML) algorithm-based techniques are used for credit risk analysis in small and medium-sized enterprises (SMEs). ML algorithms are mainly used to improve the accuracy range in the analysis process [15]. A classification and regression tree (CART) algorithm is used for risk classification and analysis. A classification technique is used in CART to analyze the relevant features for risk analysis [16]. The CART algorithm evaluates the risks and classifies the exact types of risks for the further development process [17]. The CART algorithm trains the datasets to differentiate, providing knowledge of market fluctuations. A back propagation (BP) neural network algorithm-based analysis method is also used for credit risk analysis [6], [18]. The BP algorithm detects both default and non-default risks presented in enterprises. The detected risks are used to identify the behavioural features of enterprises. The BP algorithm-based technique improves risk analysis accuracy, increasing small enterprises' robustness [12], [19]. Even though ML models may provide very precise risk evaluations, it is critical to comprehend how these findings affect financing decisions, portfolio management, and business planning. The trade-offs between model clarity and complexity must be thoroughly investigated since each ML approach provides varying degrees of interpretability. The specific factors impacting risk may be better understood by stakeholders using a more interpretable model, such as logistic regression, rather to a complicated neural network, which may provide great accuracy. A thorough evaluation is required to demonstrate how each method's results may play a role in micro-and medium-enterprise financing for business growth, risk mitigation methods, and credit allocation decisions.

The major contributions are listed below:

- i. The design and introduction of a Consistent Assessment method for financial credit risk identification and analyzing the market risks over it
- ii. A validation based on differential investment and lending knowledge for preventing stagnant organization developments using linear learning
- iii. Wide data analysis with the real-time accumulated data over varying quarters and years using the influencing factors

- iv. A performance assessment that is exclusive to the proposed method using specific methods and metrics with variant considerations.

II. RELATED WORKS

The related works section summarises a few works that various authors have proposed. The summary in Table 1 provides information on methods, features, techniques, and their findings. The table is presented below.

Maloney et al. [26] proposed a machine learning (ML) approach for predicting repayment in low-credit borrowers. The main aim of the method is to testify to the viability and usefulness range of credits. A risk data analysis technique is used here to analyze the key variables for the prediction process. The analysis technique reduces energy consumption in the risk classification process. The proposed approach improves the overall performance level of the organizations.

Abedin et al. [27] introduced a weighted synthetic minority oversampling technique (WSMOTE) using ensemble learning for small business risk prediction. WSMOTE predicts default and non-default risks presented in transaction and production processes. The WSMOTE approach is mainly used to classify the minority risk range of the organizations. The WSMOTE maximizes risk prediction accuracy, improving small businesses' robustness.

Zhao and Li [28] designed a new credit risk assessment method for small and medium-sized enterprises (SMEs). Backpropagation (BP) and support vector machine (SVM) algorithms are used to explore SMEs' financial impacts. BP neural network algorithm is mainly used here to identify the important components of supply chain financing. The BP algorithm trains the datasets and gathers relevant information for the risk prediction process. Experimental results show that the designed method increases the effective range of SMEs.

Muñoz-Izquierdo et al. [29] introduced a new machine-learning approach for credit rating assessment for enterprises. Extended audit reports are used to produce the necessary data for assessment. The key audit matter (KAM) is included in audit reports that provide optimal features for risk identification and classification processes. The introduced approach maximizes the accuracy of risk assessment, enhancing enterprises' feasibility level.

Nourmohammadi-Khiarak et al. [30] proposed a hybrid method using a meta-heuristic algorithm for feature classification in credit risk assessment. A modified fuzzy min-max classifier is used to classify the types of risks based on features. The feature selection technique is used here to identify the optimal features and variables for further processes. Compared with other methods, the proposed method improves the realisability and robustness range of the risk assessment process.

Xin et al. [31] suggested the Decision Tree Algorithm (DTA) for Enhancing Bank Credit Risk Management. This research aims to examine how the C5.0 decision tree algorithm may be used to enhance the way banks handle credit risk. To improve their credit risk management

TABLE 1. Summary of previous works.

Author	Method	Features	Technique used	Findings
Han et al. [20]	A new credit risk assessment model for peer-to-peer (P2P) network lending platforms.	The main aim is to identify the credit problems and risks presented in the platform.	A fuzzy cognitive map (FCM) is used in the model to address the risks.	Improves the effectiveness level of the risk assessment process.
Han et al. [21]	A semi-structured learning classification method for corporate credit anomaly detection.	Estimates the exact cause of anomalies in enterprises.	A generalized additive logistic regression model is used here to train the samples for assessment.	Increases the accuracy of the anomaly detection process.
Zhang et al. [22]	A firefly algorithm supports vector machine (FA-SVM) for supply chain finance (SCF) credit risk assessment.	FA-SVM identifies the risks based on an index that contains optimal data for the assessment process.	FA is mainly used here to evaluate the index of risks.	Increases the robustness of the SCFs.
Zhou et al. [23]	A Laplacian quadratic surface optimal margin distribution machine for credit risk assessment.	The actual goal is to reduce the latency in non-linearity risks in organizations.	Geometric information is used in the method to identify the margin distribution level of risks.	Enhances the performance range of small enterprises.
Lappas et al. [24]	A machine learning (ML) approach for credit risk assessment process.	ML approach predicts the power of risks that reduce the complexity of the assessment process.	A genetic algorithm with feature selection is implemented in the approach that identifies the necessary risk features.	Improves the effectiveness level of the prediction process.
Song et al. [25]	A loan default prediction method for medium and micro enterprises.	It maintains the imbalanced performance range in the classification process.	A multi-objective ensemble learning scheme is used here to detect the default risks.	Maximizes the accuracy of the classification process.

strategies, banks should move away from subjective judgment and toward objective analysis when identifying risks, and they should also move away from qualitative measurement and toward quantitative risk assessment and control. The author developed a C5.0 decision tree model to forecast high-risk bank loans using Center for Machine Learning and Intelligent Systems data. An error rate of 14.9% was revealed by evaluating the model’s performance using multiple measures, including a confusion matrix. The research shows that decision tree models may use important information like checking and savings accounts to make credit risk evaluations far more efficient and accurate.

Mahbobi et al. [32] proposed the artificial and deep neural networks (ADNN) for Credit risk classification. Using oversampling and undersampling techniques, including the synthetic minority oversampling technique (SMOTE), SVM SMOTE, random undersampling, and ALL-KNN, the suggested approach employs a 30,000 unbalanced dataset to enhance the accuracy of default payment prediction. With a cross-entropy loss of just 0.028, the findings show that the SVM using the ALL-KNN sampling method can reach 98.6 per cent accuracy. This study provides an improved understanding of how neural networks perform when coupled with resampling approaches by means of precisely implemented neural networks and neurons that make up the suggested algorithm. When applied to real-world scenarios with customers classified as non-default payers, the approach and algorithm offered here allow for more precise prediction of credit risk ratings.

Abedin et al. [27] recommended the weighted synthetic minority oversampling technique (WSMOTE-ensemble) for class-imbalanced small business credit risk prediction. To ensure the produced synthetic examples are resilient and

variable and to reduce the small business class-skewed limitations associated with default and non-default instances, the suggested ensemble classifier combines WSMOTE and Bagging with sampling composite mixes. The initial dataset for this research consisted of 3,111 records retrieved from a commercial bank in China that focused on small businesses. A multilayer experimental environment was created for unusual events by comprehensively analyzing data-modelling situations with highly skewed data. According to this research, when evaluated using the right metrics, the WSMOTE-ensemble model’s random forest classifier offers a decent compromise between the default and non-default classes’ performance. The minority class’s accuracy was 15.16 percentage points higher after using the ensemble solution than its rivals.

Wang et al. [8] discussed the machine learning techniques (MLT) for forecasting SMEs’ credit risk in supply chain finance with a sampling strategy. The author used financial data, operational data, innovation data, and negative events as predictors to create new forecast models for predicting the credit risk of small and medium-sized enterprises (SMEs) in China. These models were based on machine learning techniques and used an imbalance sampling strategy. To better anticipate the credit risk of SMEs in SCF, the empirical findings reveal that financial-based information, such as TOC and NIR, is the most relevant and that multi-source information fusion is important. Further, we introduce the different processes of important credit risk indicators for SMEs by partial dependency analysis, utilizing the favoured CSL-RF model, which expands cost-sensitive learning to a random forest.

Due to several important challenges, credit risk assessment for micro- and medium-sized businesses remains

TABLE 2. Financial/ credit statistics of the considered SME.

Quarter/ Year	Outstanding	Transactions	Growth Rate	New Credit	Impact Factor
I/2017	518	258	10.56	Yes	3
II/2017	602	315	12.08	Yes	4
III/2017	589	289	11.58	No	3
IV/2017	520	274	10.58	Yes	3
I/2018	627	329	12.8	Yes	4
II/2018	618	214	12.05	Yes	3
III/2018	489	158	11.56	Yes	2
IV/2018	410	147	10.25	No	2
I/2019	583	296	8.6	Yes	3
II/2019	420	187	7.58	No	3
III/2019	478	197	6.32	Yes	3
IV/2019	563	289	5.14	Yes	4
I/2020	374	-46	-28.5	No	1
II/2020	398	-12	-10.2	No	1
III/2020	412	158	-1.25	No	2
IV/2020	474	147	-1.58	No	1

largely unexplored in the research community. Although many advanced models for machine learning are effective, they might be difficult for non-experts or smaller institutions to employ due to their complexity and the extensive technical skills they demand. Additionally, many of these models are “black boxes,” making them difficult to understand and use, decreasing confidence in decision-making. A major challenge is the variable performance in various scenarios when dealing with unbalanced data, where default instances are less common than non-defaults. Many strategies intended for bigger companies could not work for smaller ones since their finances are more unpredictable. Lastly, important for ever-changing financial environments, real-time decision-making is lacking. Credit risk assessment for smaller businesses may be more successful if these gaps were filled with models that were easier to understand, more scalable, and more accessible.

Micro and medium enterprises heavily rely on small finance investments and returns to ensure smooth and sustained operations. However, due to inadequate planning and recovery, credit-based investments and lending often lead to financial setbacks. This article aims to examine the adverse impacts of credit risk on these enterprises and propose a solution. This article introduces the Consistent Assessment Method (CAM) using Linear Learning Analysis (LLA) to address the common challenges associated with credit investments and lending.

III. DATA COLLECTION

We utilize the data provided in (<https://www.centralbank.ie/statistics/data-and-analysis/credit-and-banking-statistics/sme-large-enterprise-credit-and-deposits>) for financial credit risk assessment using CAM-LLA. The data concerns the credit & deposit statistics of small and medium-sized enterprises for analysis. Nearly 15 enterprise types are accounted for 12 to 16 quarters of their financial outcomes. The classifications are performed as transactions and outstanding; the outstanding is further classified as non-recoverable and prolonged. We filter out the financial statistics of the “food and beverage” manufacturing SMEs from the data for analysis.

This process provides necessary Financial/ credit statistics in the following tabulation.

The above data is extracted from the statistics provided at an average rate observed between 2017 and 2020. The financial growth and credit variations are analyzed depending on the available data. Therefore, this data can classify non-recoverable and prolonged investments (Table 2).

IV. CONSISTENT ASSESSMENT MODEL USING LINEAR LEARNING ANALYSIS FOR CREDIT RISK ASSESSMENT

The proposed method aims to identify non-recoverable and prolonged investments that create financial burdens across different financial quarters. CAM provides insights into credit risks amidst market fluctuations by identifying risky investments or lending. It aims to mitigate the risks associated with these activities and ensure more stable financial processes for enterprises. The learning process involves two linear analyses: credit increment and financial development. These analyses are conducted quarterly, considering the investment and its subsequent return over the expected period. Through linear learning, this approach identifies inconsistent financial variations between investments and credit, taking into account fluctuations and instances where recovery is unlikely. This method facilitates more effective financial revisions by identifying and classifying the variants between financial improvement and risks during different quarters. It enables enterprises to make informed decisions based on a better understanding of credit risks concerning market fluctuations and non-recoverable instances. In Fig. 1, the proposed method is illustrated.

In the credit risk assessment of medium and micro enterprises, gathering information about their investments is crucial to verify further processes. This information includes details about the types of investments, and it is important to assess the scope of these investments and the industries or sectors they are associated with. Additionally, information about the terms and conditions of the investments, including repayment schedules and collateral, is vital. This data allows credit risk assessors to evaluate the potential risks and rewards

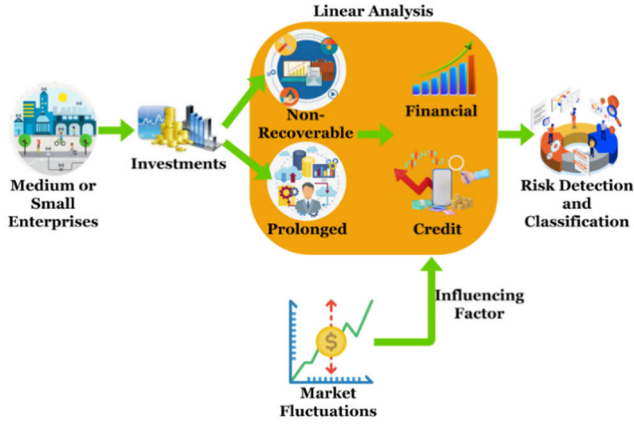


FIGURE 1. Illustration of the proposed CAM-LLA.

associated with the investments and determine the likelihood of repayment and profitability. Gathering accurate and comprehensive information about the enterprises' investments is essential for conducting a thorough credit risk assessment. Information about the investments is essential for further linear analysis of non-recoverable and prolonged investments. The process of analyzing the investments of the medium and micro enterprises in the linear analysis procedure is explained by the following equation (1) given below:

$$\left. \begin{aligned} \frac{d}{dt}a(t; \alpha) &= f(a, \alpha) \\ a(0) &= a_0 \\ a &= (a_1, \dots, a_d) \\ \alpha &= (\alpha_1, \dots, \alpha_d) \\ f(\alpha, a) : \alpha &\rightarrow a(t) \end{aligned} \right\} \quad (1)$$

where a is denoted as the investments of the medium enterprises, f is represented as the investments of the micro-enterprises, t is denoted as the payment schedules of the enterprises, α is represented as the verification operation of the investments, d is denoted as the enterprises' associated sectors. Determining the non-recoverable investments from the medium and micro enterprises investments is done through linear analysis. This approach evaluates the financial performance and risk factors associated with these investments. It helps identify the portion of investments that may not be recovered due to unfavourable circumstances. The linear analysis provides insights into the probability of investment recovery by analyzing historical data and trends. It considers market demand, competition, financial stability, and industry-specific risks. The analysis also considers external factors such as regulation changes, stability, and technological advancements that may impact investment recovery. Using linear analysis, medium and micro enterprises make informed investment decisions and mitigate potential losses. It assists in identifying areas where additional measures or adjustments may be needed to improve the chances of recovering investments and achieving sustainable growth. The process of determining the non-recoverable investments from the micro and medium enterprises'

investments is explained by the following equations (2) given below:

$$\left. \begin{aligned} X^{(i)} &= \{X(t_n^{(i)}; \alpha^{(i)}, a_0^{(i)})\} \\ &\text{where } i = 1, \dots, N \\ \{X(t_n^{(i)}; \alpha^{(i)}, a_0^{(i)}), X(t_{n+1}^{(i)}; \alpha^{(i)}, a_0^{(i)})\} \\ &\text{where } n = 0, \dots, n^{(i)} - 1 \\ \{X|0; \alpha^{(i)}, a_0^{(i)}, X(0; \alpha_n^{(i)}, a_0^{(i)})\} \\ &\text{where } n = 0, \dots, n^{(i)} + 1 \\ X &= \{a_j^{(1)}, a_j^{(2)}\} \\ &\text{where } j = 1, \dots, J \end{aligned} \right\} \quad (2)$$

where X is represented as the non-recoverable investments, i is denoted as the historical data, j is represented as the financial performance. These are investments that cannot be retained temporarily or permanently. This is analyzed using linear analysis, which helps further risk detection and classification. In the credit risk assessment, linear analysis is crucial in determining non-recoverable investments. From the data provided, the initial classification analysis for i is illustrated in Fig. 2.

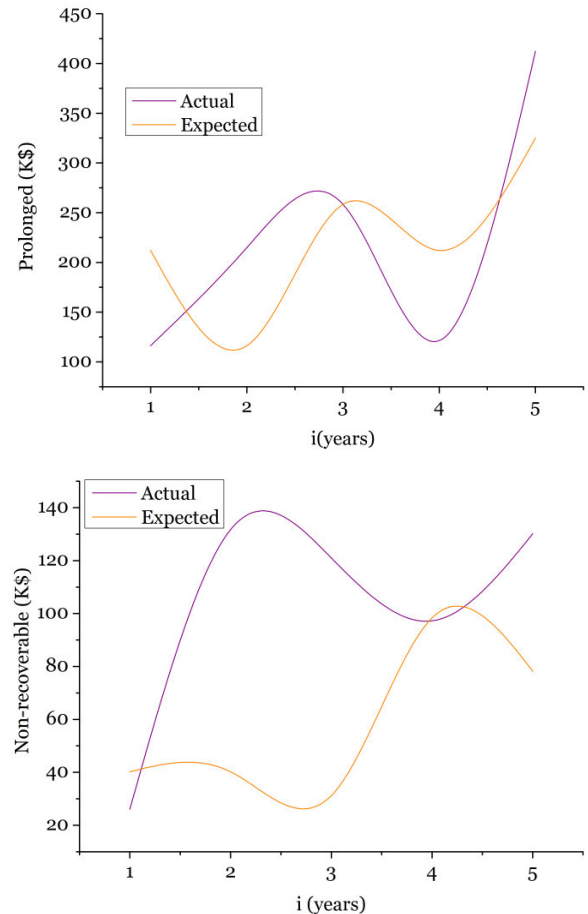


FIGURE 2. Initial classification of i (Actual and Expected).

The classifications are performed using the negative value from Table 2 data. In this process, the values are inferred in thousand dollars for ease of understanding the valid α over the a and f are summarized accordingly for the different d . This includes X^i and $(X|o)$ classifications for preventing further risks. The available data represent negative and positive remunerations in the classification process. The slide increment above shows a risk chance due to credits. Therefore, the risk prevention process (other than new credits) is implemented for enterprise management (Fig. 2). It involves evaluating various factors such as financial indicators, market conditions, and borrower characteristics to assess the creditworthiness and potential risk associated with investments. By analyzing these factors through a linear analysis approach, lenders estimate the likelihood of recovering their investments and make informed decisions regarding loan approvals, interest rates, and risk mitigation strategies. This helps ensure a more accurate credit risk assessment and enables lenders to manage their portfolios effectively. These non-recoverable investments regarding their portfolio determination procedure are also explained by the following equation (3) given below:

$$\left. \begin{aligned} \hat{Z} &= \begin{bmatrix} I_t & 0 \end{bmatrix} \\ a_1 &= [\hat{I} + \hat{N}] [a^{(i)}] \\ a_2 &= a^{(i)} + \hat{N}(x, \alpha, a) \\ Z_j^{(1)} &\rightarrow (x, \alpha, 0) \\ Z_j^{(1)} &\rightarrow (x^{(j)}) \\ \hat{N}(Z, \alpha, a) &\approx n(Z(j)) \end{aligned} \right\} \quad (3)$$

where Z is represented as the investors' portfolios for the investing operation, now the prolonged investments are identified by the linear analysis in the credit risk assessment process. In the credit risk assessment process, when it comes to medium and micro enterprises' investments, prolonged investments refer to those held for an extended period before being recovered. These investments typically involve longer repayment terms or have higher uncertainty regarding recovery. Assessing prolonged investments requires a comprehensive analysis of various factors. Lenders must evaluate these investments' financial stability, cash flow patterns, and industry-specific risks. They also consider the borrowers' repayment capacity, analyzing their historical performance and projected future earnings. Furthermore, lenders assess the market conditions, economic outlook, and any external factors that could impact the recovery of prolonged investments. This helps ensure a balanced approach to managing credit risk while supporting the growth and development of medium and micro enterprises. The process of evaluating the prolonged investments from the medium and micro enterprises' investments is explained by the following equation (4)

given below:

$$\left. \begin{aligned} Y &= Xn \\ \Rightarrow X^T n &= X^T Y \\ \Rightarrow n &= (X^T Y)^{-1} X^T Y \\ \text{where } (Y(t) &= Xn(t) \\ Y &= n_1.x_1 + n_2.x_2 + n_3.x_3 + \dots \\ X^i &= (x_1, x_2, x_3, \dots, x_d^n, \dots, 1) \end{aligned} \right\} \quad (4)$$

$$n^i = \begin{bmatrix} n_1 \\ n_2 \\ \vdots \\ n_n \\ Y \end{bmatrix}$$

where Y is denoted as the prolonged investments, these prolonged investments can be regained even after the long-term gap. In certain cases, prolonged investments can still be regained even after a long-term gap in the credit risk assessment. Factors such as improved market conditions, favourable economic changes, or the implementation of effective recovery strategies contribute to the eventual recovery of these investments, even after an extended period. The process of regaining the prolonged investments from the micro and medium enterprises is explained by the following equation (5) given below:

$$\left. \begin{aligned} q(t, x) &= -\partial t x(t, a) + (t, x) - 1 \\ x_0(t, a) &= x_n \left[Z \begin{bmatrix} t \\ x \end{bmatrix} + y \right] + x \\ q_0(t, x) &= -Z(x(t)) \\ &= -Z_1(x(t_i)) \\ &= -Z_1(x(t_j)) \end{aligned} \right\} \quad (5)$$

where $i = 1, 2, \dots, n; \quad j = 1, 2, \dots, n$

where q is denoted as the regaining operation of the prolonged investments, the financial development and credit increment are identified using linear analysis from the determined non-recoverable and prolonged investments. Determining non-recoverable and prolonged investments using linear analysis in the credit risk assessment process has implications for financial development. The identified and regained investments from the credit risks are analyzed using the previous j . However, this j experiences both downfall and appraisal between the observed quarters; the analysis is presented in Fig. 3.

Fig. 3 presents a parallel analysis of q and $j \forall$ appraisal (first two) and downfall (last two) factors. In the validating process, if q is high, then j is high. This is statistically correlated with the Table 2 data over the different outcomes. If a change in multiple $Y = X_n$ occurs, then the validation relies on n^i . Contrarily, if no improvements are observed, then linear analysis for $q_o(t, x)$ satisfying $-Z(x(t))$ is validated. This validation requires no new additional information from the previous

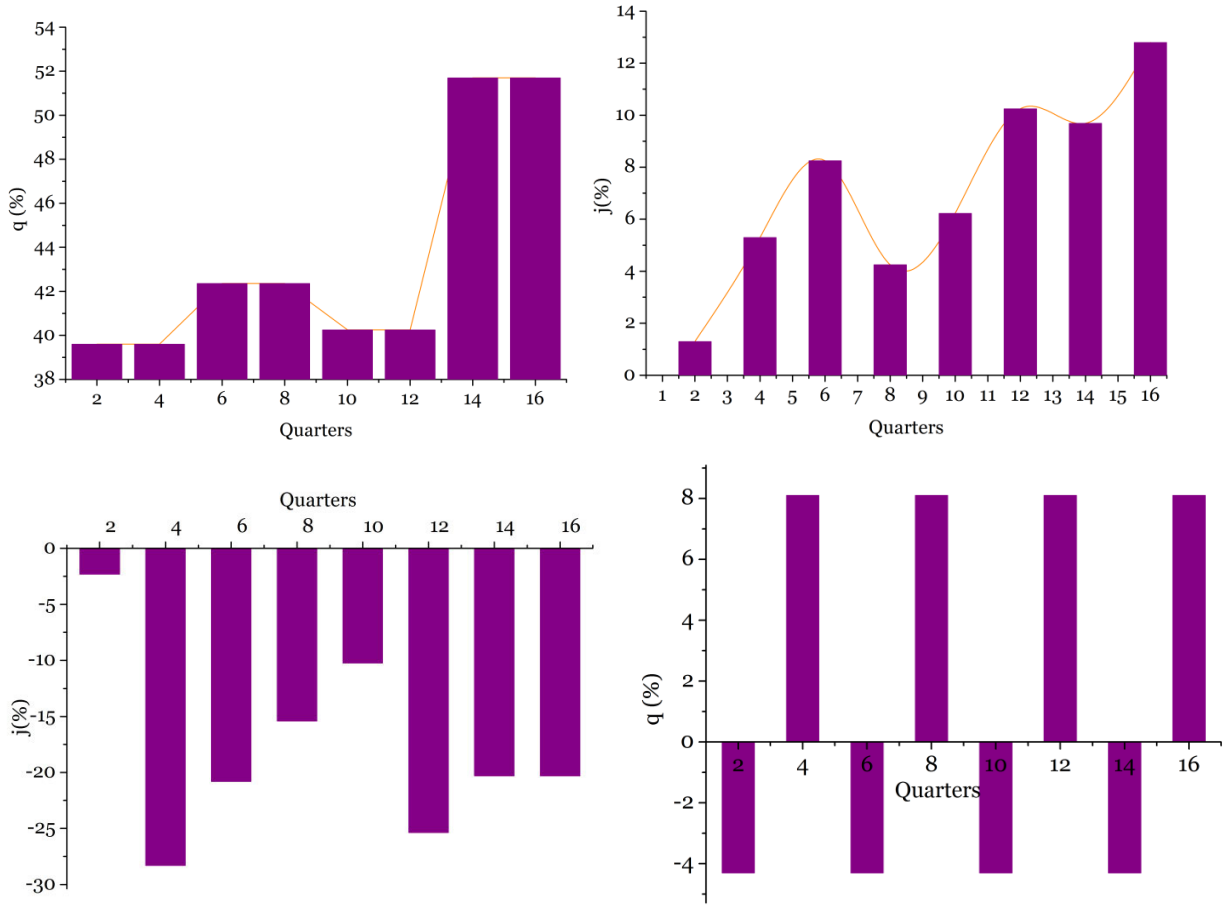


FIGURE 3. Analysis of j based on downfall and appraisal.

quarter. Therefore, the consecutive α for linearity/ variation is performed using the learning paradigm. Its output decides the new improvement decisions over the available (further) statistical data. Once these investments are identified, financial development takes appropriate measures to manage and mitigate the associated risks. Lenders adjust their loan portfolios, pricing strategies, and risk management practices by quantifying the non-recoverable and prolonged investments. This helps maintain a healthier lending environment and protects the financial stability of the enterprises' investments. This, in turn, promotes a more balanced and sustainable approach to financial development. Additionally, identifying non-recoverable and prolonged investments helps lenders make informed decisions. This ensures that appropriate provisions are set aside to cover potential losses, safeguarding the overall financial health. Using linear analysis in the credit risk assessment process contributes to financial development by enhancing risk management practices, optimizing resource allocation, and fostering a more stable and resilient lending environment. The process of determining the financial development from the non-recoverable and prolonged investments is explained by the following equations (6) & (7)

given below:

$$\left. \begin{aligned} f\left(x; \frac{\partial a}{\partial x_1}, \dots, \frac{\partial a}{\partial x_n}, \frac{\partial^2 a}{\partial x_1 \partial x_1}, \dots, \frac{\partial^2 a}{\partial x_1 \partial x_n}\right) \\ G(x, a) = 0 \\ G(x, a) = \frac{1}{T_n} \sum_{t=1}^T \left| (\hat{a}, \alpha) \right|^2 \\ G^n = \frac{\partial |n|}{\partial x_1, \dots, \partial x_n} \end{aligned} \right\} \quad (6)$$

where $n = 1, 2, \dots, nt - 1$

$$\left. \begin{aligned} \frac{\partial G}{\partial t} &= \frac{\partial a}{\partial x_1} + \frac{\partial^2 a}{\partial x_1 \partial x_n} \\ \frac{\partial y}{\partial x} + y(x) &= \int_0^1 y(t) dt \\ \frac{dx}{dt} &= \gamma(y - x) \\ \frac{dy}{dt} &= x(\gamma - z) - y \\ \frac{dz}{dt} &= xy - \gamma z \\ \frac{dy}{dt} + y(x) &= \int_0^1 y(t) dt = 1 \\ G_x(x, 0) &= G_y(x, 0) = y^{-x} \\ G_x(0, t) &= G_y(0, t) = 1 \\ G_x(1, t) &= G_y(1, t) = 0 \end{aligned} \right\} \quad (7)$$

where G is denoted as the financial development of the investments after determining the non-recoverable and prolonged investments, γ is denoted as enhancing the risk management, which helps increase the financial development. Now, the credit increment is obtained based on market fluctuations. It acts as the influencing factor for the credit increment procedure. Market fluctuations play a significant role in the credit increment process when using the linear analysis technique in credit risk assessment. Linear analysis helps assess the impact of market fluctuations on credit risk by analyzing historical data and identifying trends. The fluctuations (variations) and development observed based on j and d are analyzed in Fig. 4. This analysis relies on the data furnished in Table 1.

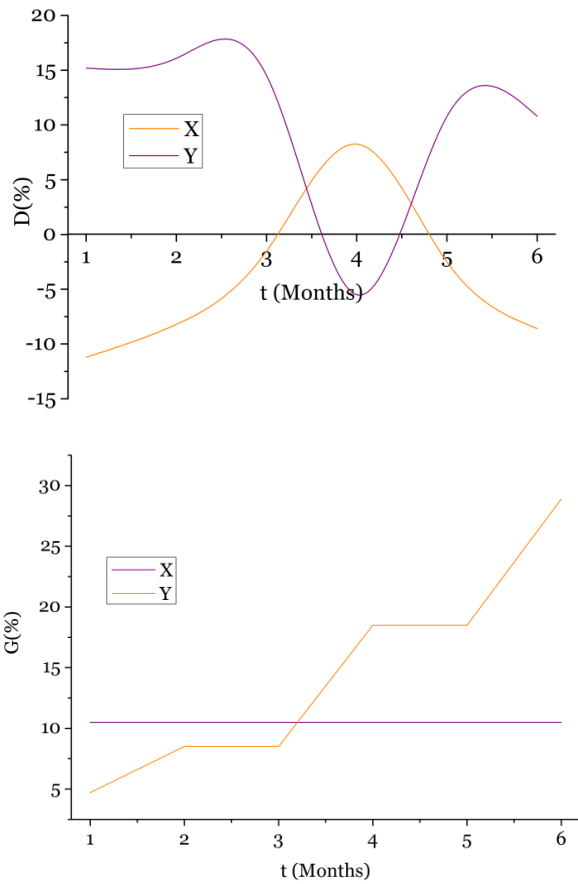


FIGURE 4. Fluctuation and development analysis (X, Y).

The D analysis uses the observed market impact on sales and stagnancy. The variation between the negative to positive or vice versa migrations from Table 2's data is used for analyzing multiple $\frac{\partial y}{\partial x}$ and $\frac{dy}{dt}$ deviations. These deviations are observed from the varying quarters and t experienced in any quarter. This is revised for $G_x(x, 0)$ for multiple X and Y variations. The change in linearity is verified using the above differentiations for preventing G failures (Fig. 4). A fluctuation in interest rates, exchange rates, commodity prices, or stock market performance affects borrowers' repayment capacity and overall credit risk. By incorporating these

factors into the analysis, lenders adjust their credit policies or collateral requirements accordingly to mitigate the risks associated with market fluctuations. This helps ensure a more accurate credit risk assessment and enables lenders to make informed decisions in the credit increment process while considering the dynamic nature of the market. The process of analyzing the market fluctuations for the credit increment process is explained by the following equations (8) & (9) given below:

$$\left. \begin{aligned} D &= \alpha_1 y_1 + \dots + \alpha_n y_n \\ \alpha &= \begin{bmatrix} \alpha_1 \\ \vdots \\ \alpha_n \end{bmatrix} \in D^n \\ D &= (d_1, \dots, d_n) \\ \gamma(D_j) &= \alpha_{ij} x_n + \dots + \alpha_{ij} x_{ij} \\ &= \sum_{i=1} \alpha_{ij} x_n \\ G\gamma(i, j) &= \alpha_{ij} \\ \text{where } i &= 1, 2, \dots, n; \quad j = 1, 2, \dots, n \end{aligned} \right\} \quad (8)$$

$$\left. \begin{aligned} x_\varphi &= [\alpha_1, \alpha_2, \alpha_3] \\ x_1 &= [-\alpha \quad -\varphi] \\ y &= (y_1, \dots, y_n) \\ \tilde{y} &= (\tilde{y}_1, \dots, \tilde{y}_n) \\ Z &= (Z_1, \dots, Z_n) \\ \tilde{Z} &= (\tilde{Z}_1, \dots, \tilde{Z}_n) \\ \varphi(y) &= Z_1 - Z_2 + 3Z_3 - Z_4 \end{aligned} \right\} \quad (9)$$

where D is denoted as the market fluctuations, φ is denoted as the market fluctuations that act as the influencing factors for enhancing the credit increment process. The credit increment process now depends on market fluctuations after determining non-recoverable and prolonged investments using linear analysis. The credit increment process considers market fluctuations after identifying non-recoverable and prolonged investments through linear credit risk assessment process analysis. The linear analysis evaluates the impact of market fluctuations on credit risk, such as interest rates, exchange rates, and commodity prices. The linear analysis learning process is illustrated in Fig. 5.

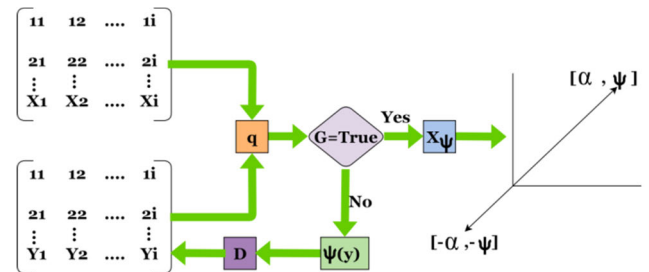


FIGURE 5. Linear analysis process for financial and credit improvements.

In the linear analysis, the (X_i, Y_i) combination inputs are used for identifying q . The aim is to validate if G is true/false based on j from the previous quarters. In this process, if G is true/false, based on j from the quarters. In this process,

if G is true, then $X_\psi \forall (\alpha, \psi)$ is verified for its improvement as in Fig. 4 (without stagnancy). If the linearity is pursued, then consecutive (X, Y) is used for improving γ . On the contrary, if G is false, then D 's impact is validated across multiple $(X, Y) \in i$. Thus, the change in $[-\alpha, -\psi]$ to $[\alpha, \psi]$ if experiences any changes the variations in financial and credit development are observed (Refer to Fig. 5). This information guides lenders in adjusting credit strategies, interest rates, and collateral requirements. This approach enhances the lender's ability to respond to changing market conditions and maintain a balanced approach to credit risk management while promoting financial stability and growth. The process of enhancing the credit increment is explained by the following equation (10) given below:

$$\left. \begin{aligned} \phi : G \rightarrow W \\ x \mapsto \alpha + \phi(x) \\ \phi * y := \phi y + \phi + y \\ \text{where } \phi, y \in G/\{-1\} \\ \phi * \phi = G_n \\ [G \setminus \{-1\}, *] \rightarrow [G_n(\alpha_{ij})] \\ \bar{Z} = \{x \in Z \mid x - n = 0\} \\ = \{x \in Z \mid \alpha \in Z : (x - n = n.\alpha)\} \end{aligned} \right\} \quad (10)$$

where ϕ is denoted as the credit increment operation, the credits are enhanced depending on market fluctuations. Identifying investments or lending activities that fall into non-recoverable and prolonged investments provides valuable insights into credit risks, particularly in the face of market fluctuations. It allows lenders to understand these investments' potential risks and challenges during market volatility. This knowledge helps lenders navigate the credit landscape more effectively and proactively manage potential risks stemming from market fluctuations. The credit increment process based on the market fluctuation is explained by the following equation (11) given below:

$$\left. \begin{aligned} P_n = \{\bar{0}, \bar{1}, \dots, \bar{n-1}\} \\ \bar{x} \oplus \bar{y} = \bar{x} + \bar{y} \\ \bar{x} \oplus \bar{y} = \bar{x} \times \bar{y} \\ P = \left\{ \begin{bmatrix} 1 & x & z \\ 1 & 1 & y \\ 0 & 0 & 1 \end{bmatrix} \in G^{n \times n} \mid x, y, z \in \alpha_{ij} \right\} \\ Px = G_{ij} \end{aligned} \right\} \quad (11)$$

where P is represented as the analyzing procedure of the potential risks in the credit risk assessment process, the risk detection process is an essential component of credit risk assessment following the linear analysis of financial development and credit increment. By combining these analyses, lenders detect various risks, including credit default or market risk. The credit increment analysis is analyzed based on D and ϕ in Fig. 6.

In Fig. 6, the ϕ in terms of thousand \$ is validated from the information provided in Table 2. This analysis is combined with X to verify its linearity. Therefore, the change in linear improvements due to $[-\alpha, -\psi]$ to $[\alpha, \psi]$ is used

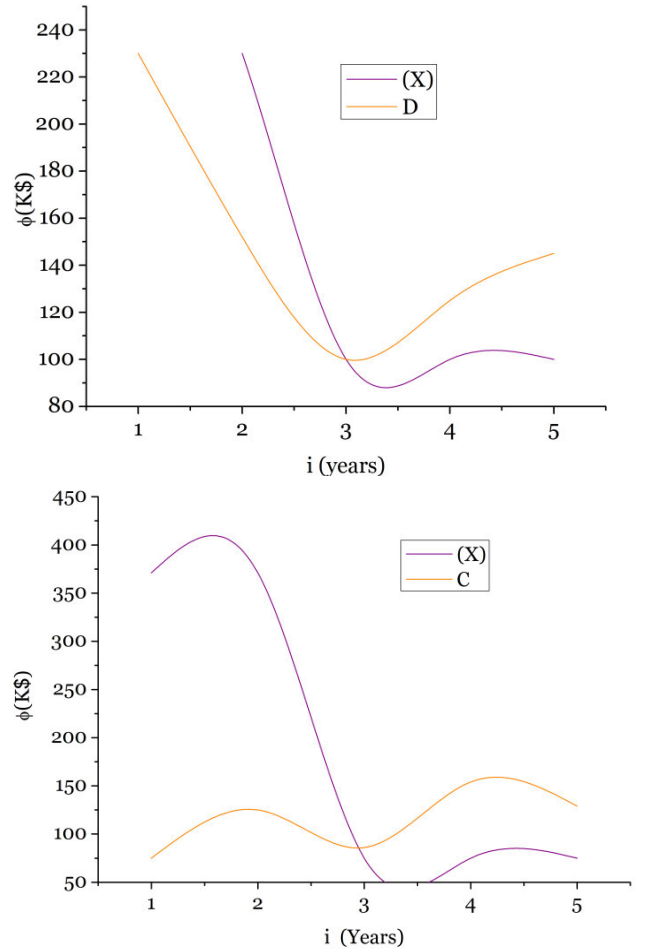


FIGURE 6. Credit increment analysis based on D and ϕ .

for identifying the variations. In this process, the demanding features are the D and ψ that are directly proportional. Hence, the changeable factors are suppressed using X_i and Y_i (independently) for $G = \text{True}$ cases. It enables lenders to take proactive measures, such as adjusting credit terms and collateral requirements or initiating risk mitigation strategies to mitigate potential losses and ensure a robust credit risk management framework. The process of detecting the risks after the two linear analyses of financial development and credit increment in the credit risk assessment is explained by the following equation (12) given below:

$$\left. \begin{aligned} \phi : H\{[x, y]\} &\rightarrow G \\ f &\mapsto \phi(f) = \int_0^1 f(x) dx \\ \phi : Z' &\dashrightarrow Z^0 \\ f &\mapsto \phi(f) = f' \\ \phi : H &\hookrightarrow H \\ x &\mapsto \phi(x) = y(t) \\ \phi : H^3 &\mapsto H^2 \\ x &\mapsto [y(t)]x \end{aligned} \right\} \quad (12)$$

where H is the risk detection operation after the linear analysis process, the classification of the detected risks to modify the investments is taking place. Applying linear learning helps analyze the inconsistent financial variations between investments and credit, considering fluctuations and non-recoverable instances. This identification allows for classifying the differences between financial improvement and risks across different quarters. The process of classification after the linear analysis is explained by the following equation (13) given below:

$$\left. \begin{aligned} \|S\| &:= \sum_{n=1} \|S_i\| \\ \|S\|_1 &= 1 \\ \|S\|_2 &= \sqrt{\sum_{i=1} S_i^2} \\ &= \sqrt{S^T S} \end{aligned} \right\} \quad (13)$$

where S is denoted as the classification process, this method helps enhance financial development and credit increments based on the fluctuations of the markets. It also helps to precisely detect the risk of modifying the investments by medium and micro enterprises. It also enhances the variance identification and classification process. This process helps increase the financial revision and reduces the time taken for the classification process. The risk detection and classification process is done precisely after obtaining the non-recoverable and prolonged investments using the linear analysis technique. Table 3 presents the H factor for the varying quarters under D , φ , and both influences.

TABLE 3. H factor analysis for different conditions.

Quarter	D	φ	H
I	✓	✓	19.9
	✓	✗	10.25
	✗	✓	9.85
	✗	✗	-12.7
II	✓	✓	18.4
	✓	✗	12.36
	✗	✓	10.54
	✗	✗	-10.5
III	✓	✓	15.6
	✓	✗	8.3
	✗	✓	5.6
	✗	✗	-8.3
IV	✓	✓	18.2
	✓	✗	9.6
	✗	✓	11.2
	✗	✗	-5.8

Table 3 (above), the H based on D and ψ is validated for the different quarters. The linear analysis verifies the variations due to $G = false$ condition in either X_i or Y_i . If both are present, the credit risk is high, and new modifications must be implemented. Considering the $\alpha \forall P$ across (D, ψ) (individual/ Combined), the successive quarters $\in t$ are analyzed. The variations are comparatively less for D and q provided (j, q) are definite (successive quarters). Thus, the prolonged factors are alone suppressed between multiple intervals to prevent non-linearity from $[-\alpha, -\psi]$ to $[\alpha, \psi]$. Therefore, credit risk assessments are optimal for making

new decisions across various quarters. Linear Learning Analysis (LLA) is a key component of the machine learning (ML) part of the Consistent Assessment Method (CAM) methodology. LLA combines the characteristics of classification models with those of linear regression. By using supervised learning methods to examine financial data from the past and form assumptions about credit risk based on trends and patterns, this method makes CAM flexible enough to tackle various risk situations. Making the risk score more precise and more accessible, CAM-LLA prioritizes interpretability, unlike many black-box models. This means that decision-makers can understand how each input information, such as financial ratios and credit history, impacts the risk score. CAM is a kind of ML that enhances decision-making via data-driven learning processes. Much like conventional ML algorithms modify their parameters and weights to reduce prediction error, it always improves its evaluation by modifying models using fresh data. Unlike more complicated non-linear models, such as neural networks, this approach is linear; thus, the relationships between the inputs and outputs are always apparent. CAM-LLA offers several benefits compared to current approaches. For example, although ensemble learning and Support Vector Machines (SVM) are two examples of classic models, they aren't always interpretable, even when they provide quite accurate results. CAM's emphasis on linear learning makes it easier for practitioners to comprehend the variables that influence risk projections, increasing confidence in the model's outcomes. Not only that, in contrast to more complicated algorithms like deep neural networks, it simplifies implementation and requires less technical knowledge. The trade-off, however, is that CAM-LLA may not be up to standard when faced with very non-linear data, while more sophisticated models might be more predictive.

V. PERFORMANCE ASSESSMENT

The performance assessment is validated using the metrics of risk detection, variation detection, variant classifications, financial revisions, and classification time. The classifications are between 5 and 60, and the financial quarters are between 1 and 15. The accompanying methods in the assessment are FA-SVM [22], WSMOTE-EM [27], and SVM-BP [28] from the related works section.

A. RISK DETECTION

Risk detection is efficacious in this process, using linear analysis to determine the state of the medium and micro enterprises' investments. High-risk detection using linear analysis involves assessing financial development and credit increments as part of the credit risk detection process. The linear analysis examines the relationship between various financial indicators and credit risk to identify potential high-risk scenarios. Credit increments refer to changes in credit behaviour, such as increased borrowing or a rise in debt levels. Linear analysis helps identify patterns and correlations between credit increments and credit risk. It considers factors

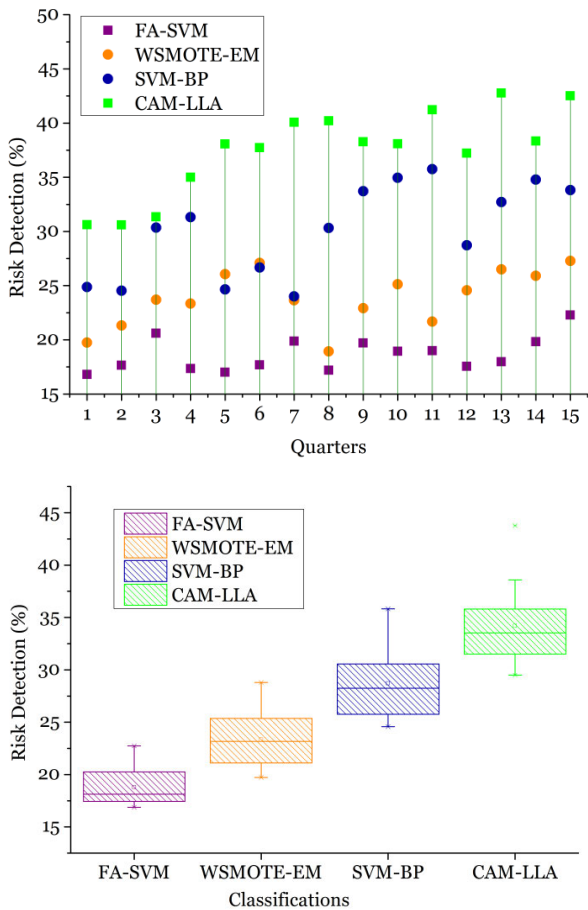


FIGURE 7. Risk detection performance assessment.

like repayment history, credit utilization, and credit scores. Lenders detect high-risk individuals or businesses with potential repayment difficulties or default risks using this approach. It enables them to make informed decisions regarding lending, setting appropriate interest rates, or implementing risk mitigation strategies to protect their interests. By this, risk detection is effective in this credit risk assessment operation (Fig. 7).

B. VARIATION IDENTIFICATION

Determining the variation is effective in using linear analysis in this credit risk assessment process. Effectively identifying variations between improvements and risks using linear analysis in credit risk assessment refers to using linear models to assess a credit opportunity’s potential benefits and drawbacks. The linear analysis examines various factors, such as financial indicators, borrower characteristics, and market conditions, to evaluate the potential for positive outcomes and the associated risks. By utilizing linear analysis outcomes, investors quantify the impact of different variables on credit risk and make informed decisions. This method helps identify the factors that contribute to improvements, such as lower default probabilities or higher repayment rates,

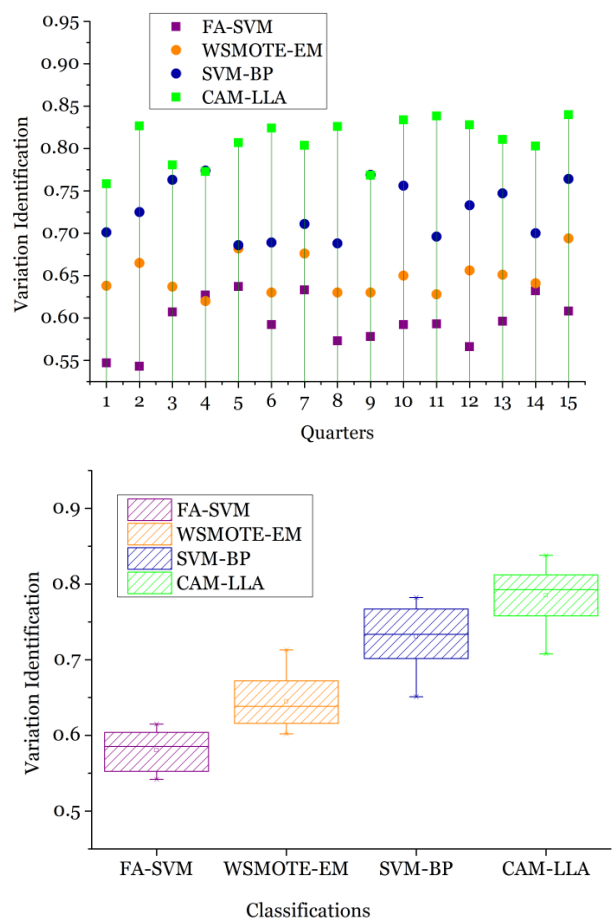


FIGURE 8. Variation identification performance assessment.

as well as potential risks, such as higher interest rates or unfavourable market conditions. By understanding these variations, lenders and financial institutions effectively evaluate creditworthiness and make well-informed decisions when extending credit to individuals or businesses (Fig. 8).

C. VARIANT CLASSIFICATIONS

The classification of the variant is efficacious in this method with the help of the output of the linear analysis process. In credit risk assessment, the output of the linear analysis process assists in achieving a better variant classification. Linear analysis helps identify the key variables contributing to credit risk by examining their relationships and impact. This output provides valuable insights for classifying variants more effectively. By analyzing the output of the linear analysis, financial institutions determine the significance and direction of each variable’s influence on credit risk. This information helps to classify variants based on their importance and contribution to overall risk. Lenders prioritize and assess risk factors more accurately with a better variant classification. They assign appropriate weights and importance to each variant, allowing for a more refined assessment of creditworthiness. Ultimately, this leads to more informed lending decisions and a more robust credit risk assessment process (Fig. 9).

TABLE 4. Performance assessment–inference.

Metrics	FA-SVM	WSMOTE-EM	SVM-BP	CAM-LLA	Inferred Results
Variant= Quarters					
Risk Detection (%)	22.28	27.28	33.81	42.509	14.72% ↑
Variation Identification	0.608	0.694	0.764	0.8398	15.11% ↑
Variant Classification	12	29	48	64	8.94% ↑
Financial Revisions (/Quarters)	5	7	9	11	6.06% ↑
Classification Time (min)	11.94	10.21	8.81	7.155	10.22% ↓
Variant= Classifications					
Risk Detection (%)	22.75	28.8	35.82	43.774	14.65% ↑
Variation Identification	0.604	0.713	0.782	0.8379	13.82% ↑
Financial Revisions (/Quarters)	5	7	10	11	5.56% ↑
Classification Time (min)	12.03	9.92	8.36	6.458	12.03% ↓

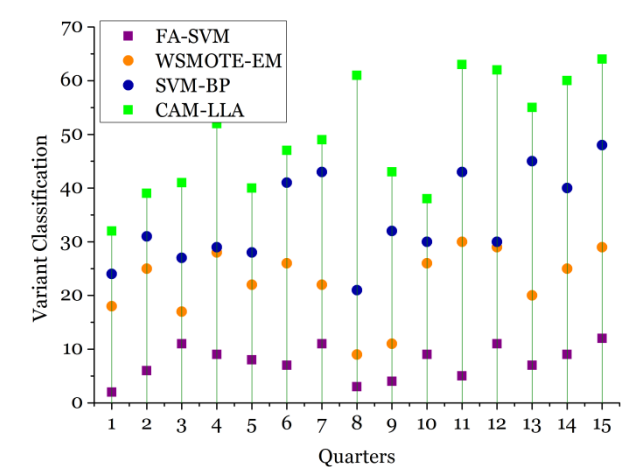


FIGURE 9. Variant classifications performance assessment.

D. FINANCIAL REVISIONS

The financial revisions are better in this credit risk assessment process by using the linear analysis technique to determine the status of the investments from the enterprises. In the credit risk assessment, linear analysis facilitates better financial revisions, particularly when dealing with non-recoverable and prolonged investments. The linear analysis allows for a systematic evaluation of various financial indicators and their impact on credit risk. Financial revisions are made to mitigate the associated risks once non-recoverable and prolonged investments are identified through linear analysis. Based on this information, appropriate adjustments are made to the credit risk assessment process. Through linear analysis, financial developments refine their credit risk assessment models and make more informed decisions regarding non-recoverable and prolonged investments. This helps mitigate potential losses and enhances the overall stability and effectiveness of the credit risk assessment process (Fig. 10).

E. CLASSIFICATION TIME

The time taken for the classification process after the linear analysis process is less after obtaining the financial development and credit increment operation. The output of the linear analysis process in credit risk assessment helps to reduce classification time, especially in market fluctuations.

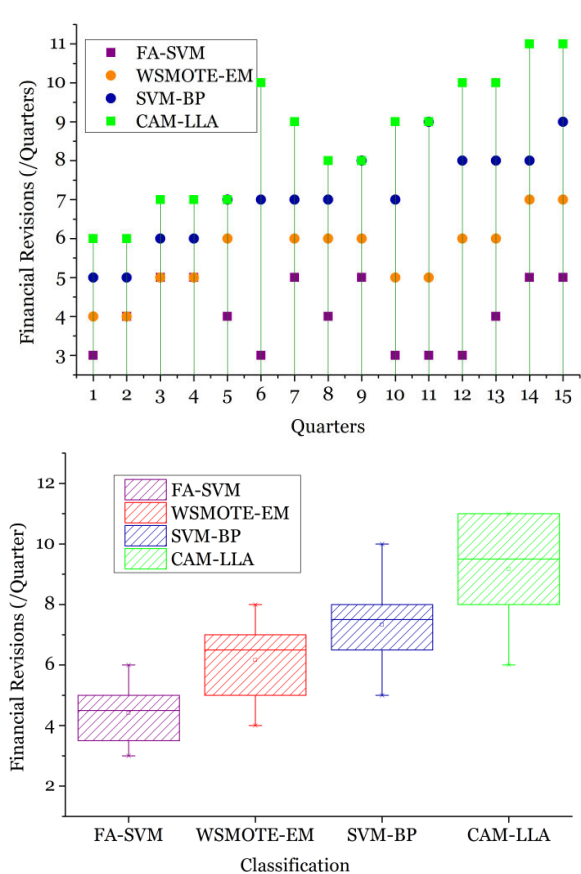


FIGURE 10. Financial revisions performance assessment.

Linear analysis helps identify the key variables and their relationships to credit risk, which is invaluable during market volatility. By analyzing the output of the linear analysis, financial development determines which variables are most sensitive to market fluctuations and have a significant impact on credit risk. This knowledge allows for a more targeted and efficient classification process. With a clear understanding of the influential factors, lenders prioritize their assessment efforts and allocate resources accordingly. This helps enhance risk management and adaptability in the face of market fluctuations, ultimately improving the overall effectiveness of the credit risk assessment process (Fig. 11). The performance assessment above is inferred from the results in Table 4.

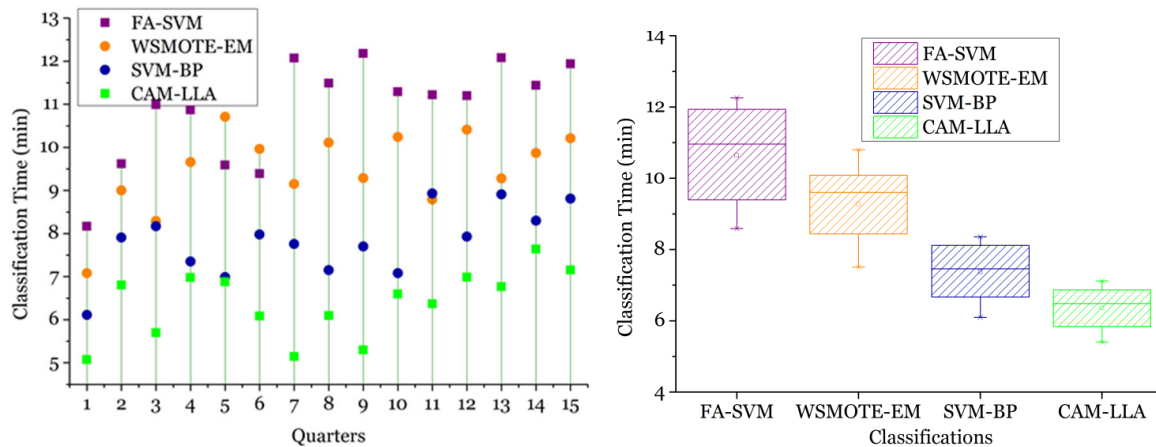


FIGURE 11. Classification time performance assessment.

A user-friendly interface that automates critical operations with minimal need for technical knowledge might ease applying the CAM-LLA approach for credit risk assessment, making it more accessible. Practitioners can connect with the system more naturally with pre-configured models and explicit data input and interpretation directions. Users may concentrate on certain parts of the procedure without being overwhelmed by the entire complexity thanks to a modular approach that breaks down the CAM-LLA technique into discrete, manageable segments. If it were simpler and more guided, its use in the real world would be substantially enhanced.

VI. CONCLUDING REMARKS

This article introduced the consistent assessment method using linear learning analysis to improve credit risk assessment and classification. The proposed method identifies the overheads in financial developments based on market variations and fluctuations. In particular, the overheads based on non-recoverable and prolonged investments are identified under multiple quarters for assessment. Linear learning is trained to identify variations in credits and financial development of organizations. Those identified variations are addressed using appropriate decisions on further investments and lending other than credits. Such a process is based on market demands and variations impacting the organization's growth in different quarters. This learning analysis process periodically identifies breaks and credit risks for revamping investment policies and rules to prevent continuous credit downfalls. Therefore, the variants are classified based on actual fluctuations observed due to non-recoverable investments that lead to further financial risks. Thus, this proposed method achieves 14.65% high-risk detection with 5.56% financial revisions for the industry considered by the data source. Datasets showing complicated, non-linear relationships may challenge the CAM-LLA approach since it is based on linear learning analysis. This limitation could reduce its ability to capture complex interdependencies among

variables when contrasted with non-linear models such as decision trees or neural networks. Making CAM-LLA more suited for high-frequency lending environments, such as peer-to-peer lending platforms or digital credit scoring systems, might be a future goal of improvements aimed at optimizing it for real-time assessments.

REFERENCES

- [1] Y. Wang, Y. Zhang, Y. Lu, and X. Yu, "A comparative assessment of credit risk model based on machine learning—A case study of bank loan data," *Proc. Comput. Sci.*, vol. 174, pp. 141–149, 2020, doi: [10.1016/j.procs.2020.06.069](https://doi.org/10.1016/j.procs.2020.06.069).
- [2] G. S. Alzamora, M. R. Aceituno-Rojas, and H. I. Condori-Alejo, "An assertive machine learning model for rural micro credit assessment in Peru," *Proc. Comput. Sci.*, vol. 202, pp. 301–306, 2022, doi: [10.1016/j.procs.2022.04.040](https://doi.org/10.1016/j.procs.2022.04.040).
- [3] X. Jin, Y. Ke, and X. Chen, "Credit pricing for financing of small and micro enterprises under government credit enhancement: Leverage effect or credit constraint effect," *J. Bus. Res.*, vol. 138, pp. 185–192, Jan. 2022.
- [4] Q. Qian, X. Chao, and H. Feng, "Internal or external control? How to respond to credit risk contagion in complex enterprises network," *Int. Rev. Financial Anal.*, vol. 87, May 2023, Art. no. 102604.
- [5] S. Xu and M. Zhang, "Research on credit risk assessment of listed companies in technology sector based on support vector machine integration," *Proc. Comput. Sci.*, vol. 214, pp. 867–874, 2022, doi: [10.1016/j.procs.2022.11.253](https://doi.org/10.1016/j.procs.2022.11.253).
- [6] Y. Hu and J. Su, "Research on credit risk evaluation of commercial banks based on artificial neural network model," *Proc. Comput. Sci.*, vol. 199, pp. 1168–1176, 2022, doi: [10.1016/j.procs.2022.01.148](https://doi.org/10.1016/j.procs.2022.01.148).
- [7] H. I. Condori-Alejo, M. R. Aceituno-Rojas, and G. S. Alzamora, "Rural micro credit assessment using machine learning in a Peruvian microfinance institution," *Proc. Comput. Sci.*, vol. 187, pp. 408–413, 2021, doi: [10.1016/j.procs.2021.04.117](https://doi.org/10.1016/j.procs.2021.04.117).
- [8] L. Wang, F. Jia, L. Chen, and Q. Xu, "Forecasting SMEs' credit risk in supply chain finance with a sampling strategy based on machine learning techniques," *Ann. Oper. Res.*, vol. 331, no. 1, pp. 1–33, Dec. 2022.
- [9] S. Shi, R. Tse, W. Luo, S. D'Addona, and G. Pau, "Machine learning-driven credit risk: A systemic review," *Neural Comput. Appl.*, vol. 34, no. 17, pp. 14327–14339, Sep. 2022.
- [10] H. A. Meressa, "Micro-and small-scale enterprises' financing preference in line with POH and access to credit: Empirical evidence from entrepreneurs in Ethiopia," *J. Innov. Entrepreneurship*, vol. 11, no. 1, pp. 1–30, 2022.
- [11] L. Wang and W. Zhang, "A qualitatively analyzable two-stage ensemble model based on machine learning for credit risk early warning: Evidence from Chinese manufacturing companies," *Inf. Process. Manage.*, vol. 60, no. 3, May 2023, Art. no. 103267.

- [12] A. Alonso-Robisco and J. M. Carbó, "Can machine learning models save capital for banks? Evidence from a Spanish credit portfolio," *Int. Rev. Financial Anal.*, vol. 84, Nov. 2022, Art. no. 102372.
- [13] L. Yu, X. Huang, and H. Yin, "Can machine learning paradigm improve attribute noise problem in credit risk classification?" *Int. Rev. Econ. Finance*, vol. 70, pp. 440–455, Nov. 2020.
- [14] K. Xu, Q. Qian, X. Xie, and Z. Zhou, "Study on the contagion mechanism of associated credit risk with corporate senior executives' alertness," *Proc. Comput. Sci.*, vol. 199, pp. 207–214, 2022, doi: [10.1016/j.procs.2022.01.026](https://doi.org/10.1016/j.procs.2022.01.026).
- [15] Y. Li and Q. Du, "The cascade effect of associated credit risk contagion among enterprises under the economic cycles," *Proc. Comput. Sci.*, vol. 214, pp. 904–909, 2022, doi: [10.1016/j.procs.2022.11.258](https://doi.org/10.1016/j.procs.2022.11.258).
- [16] Y. Lu, L. Yang, B. Shi, J. Li, and M. Z. Abedin, "A novel framework of credit risk feature selection for SMEs during Industry 4.0," *Ann. Oper. Res.*, pp. 1–28, Jul. 2022, doi: [10.1007/s10479-022-04849-3](https://doi.org/10.1007/s10479-022-04849-3).
- [17] F. Wang, L. Ding, H. Yu, and Y. Zhao, "Big data analytics on enterprise credit risk evaluation of e-Business platform," *Inf. Syst. e-Bus. Manage.*, vol. 18, no. 3, pp. 311–350, Sep. 2020.
- [18] A. Alonso Robisco and J. M. Carbó Martínez, "Measuring the model risk-adjusted performance of machine learning algorithms in credit default prediction," *Financial Innov.*, vol. 8, no. 1, p. 70, Dec. 2022.
- [19] D. Bams, M. Pisa, and C. C. Wolff, "Spillovers to small business credit risk," *Small Bus. Econ.*, vol. 57, pp. 323–352, Jun. 2021.
- [20] H. Han, Y. Yang, R. Zhang, and B. Brekhna, "FCM-based P2P network lending platform credit risk dynamic assessment," *IEEE Access*, vol. 8, pp. 195664–195674, 2020.
- [21] S. Han, "Semi-supervised learning classification based on generalized additive logistic regression for corporate credit anomaly detection," *IEEE Access*, vol. 8, pp. 199060–199069, 2020.
- [22] H. Zhang, Y. Shi, X. Yang, and R. Zhou, "A firefly algorithm modified support vector machine for the credit risk assessment of supply chain finance," *Res. Int. Bus. Finance*, vol. 58, Dec. 2021, Art. no. 101482.
- [23] J. Zhou, Y. Tian, J. Luo, and Q. Zhai, "A kernel-free Laplacian quadratic surface optimal margin distribution machine with application to credit risk assessment," *Appl. Soft Comput.*, vol. 133, Jan. 2023, Art. no. 109931.
- [24] P. Z. Lappas and A. N. Yannacopoulos, "A machine learning approach combining expert knowledge with genetic algorithms in feature selection for credit risk assessment," *Appl. Soft Comput.*, vol. 107, Aug. 2021, Art. no. 107391.
- [25] Y. Song, Y. Wang, X. Ye, R. Zaretski, and C. Liu, "Loan default prediction using a credit rating-specific and multi-objective ensemble learning scheme," *Inf. Sci.*, vol. 629, pp. 599–617, Jun. 2023.
- [26] D. Maloney, S.-C. Hong, and B. N. Nag, "Two class Bayes point machines in repayment prediction of low credit borrowers," *Heliyon*, vol. 8, no. 11, Nov. 2022, Art. no. e11479.
- [27] M. Z. Abedin, C. Guotai, P. Hajek, and T. Zhang, "Combining weighted SMOTE with ensemble learning for the class-imbalanced prediction of small business credit risk," *Complex Intell. Syst.*, vol. 9, no. 4, pp. 3559–3579, Aug. 2023.
- [28] J. Zhao and B. Li, "Credit risk assessment of small and medium-sized enterprises in supply chain finance based on SVM and BP neural network," *Neural Comput. Appl.*, vol. 34, no. 15, pp. 12467–12478, Aug. 2022.
- [29] N. Muñoz-Izquierdo, M. J. Segovia-Vargas, M.-D.-M. Camacho-Miñano, and Y. Pérez-Pérez, "Machine learning in corporate credit rating assessment using the expanded audit report," *Mach. Learn.*, vol. 111, no. 11, pp. 4183–4215, Nov. 2022.
- [30] J. Nourmohammadi-Khiarak, M.-R. Feizi-Derakhshi, F. Razeghi, S. Mazaheri, Y. Zamani-Harghalani, and R. Moosavi-Tayebi, "New hybrid method for feature selection and classification using meta-heuristic algorithm in credit risk assessment," *Iran J. Comput. Sci.*, vol. 3, no. 1, pp. 1–11, Mar. 2020.
- [31] Q. Xin, R. Song, Z. Wang, Z. Xu, and F. Zhao, "Enhancing bank credit risk management using the C5.0 decision tree algorithm," *J. Environ. Sci. Technol.*, vol. 3, no. 1, pp. 960–967, 2024.
- [32] M. Mahbobi, S. Kimiagari, and M. Vasudevan, "Credit risk classification: An integrated predictive accuracy algorithm using artificial and deep neural networks," *Ann. Oper. Res.*, vol. 330, nos. 1–2, pp. 609–637, Nov. 2023.



YUJIE ZHAO was born in Lanzhou, Gansu, China, in 1986. She received the master's degree from Central South University, China. Currently, she is with the College of Management, Shenzhen Institute of Information Technology. Her research interests include software trustworthiness, cloud computing, and big data.

...